25900751 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Discipline Specific Core Course (DSC) - MG1DSCBOT100, MG1DSCBBT100 - FASCINATING WORLD OF PLANT SCIENCES

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **10** Questions

Each Question carries **1** mark

1. What family does Quinine Tree belong to? [K] / [CO1]
2. Identify the significance of superhydrophobic surfaces [K] / [CO2]
3. How paper chromatography is typically carried out. [K] / [CO3]
4. What are the common reproduction methods in algae? [K] / [CO2]
5. Lichens are a symbiotic relationship between fungi and _____. [K] / [CO2]
6. Which is the source of Taq DNA polymerase enyme? [U] / [CO3]
7. What is the main characteristic feature of gymnosperm seeds? [K] / [CO2]
8. Illustrate different types of venations on leaves [A] / [CO3]
9. Define GIS? [K] / [CO3]

10. List the main parts of a compound microscope [K] / [CO3]

11. Who is considered as the "Father of Botany"? [K] / [CO1]

12. List any two applications of plant genetic engineering. [U] / [CO4, CO5]

[10x1 = 10]

Part B

Short Essay Type Questions

Answer any **6** Questions

Each Question carries **5** marks

13. Explain the statement - all food comes from plants. [A] / [CO1]

14. Discuss the evolutionary trends leading to the dominance of angiosperms in plant diversity. [A] / [CO2]

15. How can a plant science scholar evaluate the extend of forests and vegetation patterns of a large land area? [E] / [CO5]

16. Wtite a short note on plants living in extreme environments [K] / [CO2]

17. Comment on the importance of botanical illustrations [U] / [CO3]

18. Describe how smart devices can be used in agriculture to manage water stress? [A] / [CO5]

19. Analyse the significance of fig leaves in lemur diets. [An] / [CO2]

20. What are the differences between a TS and LS [U] / [CO3]

[6x5 = 30]

Part C

Essay Type Questions

Answer any **1** Question

Question carries **10** marks

21. Give and account on the special adaptations in plants with reference to the types to heliotropisim and pseudocopulation in orchids. [U] / [CO2]

22. Describe the different types of microtomes and discuss their specific applications. [U] / [CO3]

[1x10 = 10]